Khwaja Yunus Ali University

Lab Report

Name of the Department: **Computer Science and Engineering**

Course Code: **CSE 0613 - 1202**

Course Title: **Stracture Programing Language Lab**

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Problem no- 1

```c
#include <stdio.h>
int call = 0;
int fib(int a)
{
    call++;
    if (a == 0)
        return 0;
    else if (a == 1)
        return 1;
    else
        return (fib(a - 1) + fib(a - 2));
}

int main()
{
    int T, fib(int);
    scanf("%d", &T);

    while (T--)
    {
        int N, fibN;
        scanf("%d", &N);
        call = 0;
        fibN = fib(N);
        printf("fib(%d) = %d calls = %d\n", N, call - 1, fibN);
    }

    return 0;
}
```

Problem no- 2

```c
#include <stdio.h>
int main()
{
    int T, N;
    long long fib[61];
    scanf("%d", &T);
    fib[0] = 0;
    fib[1] = 1;
    for (int i = 2; i <= 60; i++)
    {
        fib[i] = fib[i - 1] + fib[i - 2];
    }
    for (int j = 1; j <= T; j++)
    {
        scanf("%d", &N);
        printf("Fib(%d) = %lld\n", N, fib[N]);
    }
    return 0;
}
```

Problem no -3

Write a c program to check a string is palindrome or not without using library function .

```c
#include <stdio.h>

int main() {
    char word[100];
    int i, length = 0, flag = 1;

    printf("Enter a word: ");
    gets(word);

    for (i = 0; word[i] != '\0'; i++) {
        length++;
    }

    for (i = 0; i < length / 2; i++) {
        if (word[i] != word[length - i - 1]) {
            flag = 0;
            break;
        }
    }

    if (flag == 1) {
        printf("Palindrome\n");
    } else {
        printf("Not Palindrome\n");
    }

    return 0;
}
```